Aswith Reddy Sama

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PROFESSIONAL SUMMARY

AI Engineer with **3+ years** of combined **AI/ML** and **software engineering experience**, specializing in production-grade LLM applications, multi-agent systems, and RAG. Experienced in building end-to-end AI applications with deterministic workflows and guardrails, fine-tuning transformer models with PEFT/LoRA, and developing scalable backend services with Python, FastAPI, PostgreSQL, Docker, AWS, and GCP.

PROFESSIONAL EXPERIENCE

Data Scientist | Chicago Education Advocacy Cooperative (NGO) **February 2026 – Present**

*Project: **Venue Intelligence Platform** | scikit-learn, LangChain, RAG* Chicago, USA

- **Built** a **RAG**-powered Intelligent Platform integrating internal operational data with external knowledge sources.
- Built **regression** and **tree-based** ML models using operational data for attendance, dwell time, and revenue forecasting, **integrating model predictions** with **RAG** and **LLM reasoning** to generate evidence-backed business recommendations.

AI Engineer | Catapult **May 2026 – July 2026**

*Project: **Intelligent-Multi-Tenant-Messaging-System** | LangGraph, OpenAI, FastAPI* Utah, USA

- Architected a 3-layer **LangGraph-based multi-agent AI** system automating customer service, classifying inbound messages across Upsell, Billing, After-Service, and Review with **15+ sub-domain categories** guiding **LLM decision-making** for 70+ pest-control companies.
- Engineered a **deterministic AI framework** with production **guardrails,** restricting LLMs to approved workflows while separate execution logic handled **tool calling** and PostgreSQL updates, reducing hallucinations and ensuring safe, **consistent** automation.
- Engineered an **event-driven AI pipeline** integrating LLMs, FastAPI, PostgreSQL, Twilio, and CRMs, with **microservices** separating the **backend and AI worker on GCP Cloud Run** and Cloud Tasks enabling asynchronous LLM execution.
- Collaborated with CEO to refine layered prompt and **context engineering**, passing only relevant decision context between agent layers.
- Deployed on **Cloud Run** and achieved **~95% production classification accuracy** using **GPT-5-mini hybrid reasoning.**

AI/ML Algorithm Engineering Intern | Uplifty **July 2025 – November 2025**

*Project: **Human Wellness NLP Pipeline** | LLM Orchestration, PyTorch, BERTweet* New York, USA

- **Spearheaded** the **Human Growth Index**, designed a multi-dimensional well-being measure and a **multi-agent LLM weak-labeling pipeline** annotating **100K+ texts**, routing ~8% conflicting GPT-4o-mini classifications to GPT-4.1, escalating unresolved cases to human review.
- **Fine-tuned** BERTweet with **PEFT/LoRA** on the curated dataset, achieving **0.70–0.90 F1 scores** across 27 emotion and 13 well-being categories; transformed predictions into explainable well-being score and deployed production APIs using Flask & GCP.

Software Engineer / Analyst | Unizen Technologies Pvt. Ltd. **May 2022 – December 2023**

Project: Learning Management System (LMS) Backend Platform | Node.js, Express.js, AWS India

- Developed and maintained **20+ REST APIs** using **Node.js** and **Express.js** for **core LMS workflows**, supporting 5,000+ users.
- Designed **MySQL** schemas and **backend business logic** for admissions, course management, reducing API response latency by ~20%.
- Integrated Amazon **S3** for secure document storage and automated CI/CD with AWS **CodePipeline**.

Project: Enterprise Data Pipeline | PySpark, PowerBI, scikit-learn

- Partnered with senior engineers to analyze sales, inventory data in **Power BI**, supporting business insights and demand forecasting.
- Transformed 100K+ records using **Databricks (PySpark)**, through data cleaning & feature engineering for analytics and ML workflows.
- Assisted in developing Azure Data Factory **ETL pipelines** to ingest data from SAP ERP, SQL Server, and flat files into the analytics platform.

PROJECTS

AI Engineer | ***RAG-Powered Policy Retrieval System*** | Hack Midwest **October 2025 – November 2025**

- Built a **multi-stage RAG pipeline** for health-insurance policy Q&A, transforming policy PDFs with **PyMuPDF + Docling** into **38 policy-aware chunks and 105 fine-grained internal sections** for hierarchical retrieval.
- Engineered **3-level semantic retrieval** with **768-D BGE embeddings** across narrative text, LLM-generated summaries, and internal sections, combining **MongoDB Atlas Vector Search + BM25 + RRF** to achieve **~85% Recall@3 across 50+ evaluation queries**.
- Implemented **MiniLM CrossEncoder reranking** and grounding **GPT-5-mini** responses in policy-specific conditions, exclusions, and limits.

Teaching Assistant | ***Automated Grading System*** | Illinois Institute of Technology **August 2025 – December 2025**

- **Mentored** 50+ graduate students in building **GIS-based network visualizations** of Chicago communities using **Python** and **Power BI**.
- **Built a hybrid AI grading pipeline**, processing **zip/docx files** with typed, **handwritten solutions** using **Vision LLMs**, schema-constrained extraction, and **deterministic grading rules**, routing 20% of ambiguous answers for **human review** and generating **Excel audit reports**.

SKILLS & CERTIFICATIONS

- **Languages:** Python, Java, C++, SQL.
- **AI Engineering:** LLMs, OpenAI API, Multi-Agent Architectures & Orchestration, ReAct, LangChain, LangGraph, Prompt & Context Engineering, Deterministic AI Workflows, Tool Calling, AI Guardrails, Vector DBs, RAG, Hybrid Retrieval, Cross-Encoder Reranking, MCP.
- **Deep Learning:** TensorFlow, PyTorch, NLP, Transformer Models, PEFT/LoRA, CNNs, RNNs, YOLOv3.
- **Backend & Cloud:** FastAPI, REST APIs, PostgreSQL, Node.js, Pydantic, Docker, Git, CI/CD Pipelines, AWS (S3, Bedrock), GCP (Cloud Run).
- **Certifications:** AWS Certified Machine Learning Associate, Mathematics for Machine Learning.

EDUCATION

Illinois Institute of Technology | Chicago, USA **January 2024 – December 2025**

Master of Science in Artificial Intelligence — GPA: 3.70/4.00

J.B. Institute of Engineering and Technology | Hyderabad, India **July 2019 – May 2023**

Bachelor of Technology in Information Technology — GPA: 3.57/4.00